

# Lecture 2 — Why This Object, Honestly

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## Lecture 2 — Why This Object, Honestly

### The question

Mathematics offers infinitely many shapes. Why this one?

For most of 2026 our answer was a theorem with eleven criteria and a probability: no other bounded symmetric domain satisfied all eleven, and if the criteria were independent the chance of that happening by accident was about one in a billion. It was called the Strong-Uniqueness Theorem, it has a chapter in the legacy curriculum (Vol 0, Chapter 9), and in September 2026 we audited it and took most of it down. This lecture tells you what survived, because what survived is better than what we had — smaller, sharper, and honest about where a measurement enters.

### For the reader in a hurry

Every shape in the classification comes with a small whole number attached, called its multiplicity — think of it as how many ways the shape branches at each step toward its edge. Quarks come in three colours. There is exactly one shape in the entire classification whose multiplicity is three. That is why we study this one. And here is the honest part: the fact that colour *is* that number — rather than some other three — is something we assume, and we have proved that the shape itself cannot tell us why. So the shape is not forced on us by mathematics. It is picked out by one measured fact, and we say which.

### What went wrong with eleven criteria

The eleven-criterion argument had three faults, and they are worth naming because they are the faults such arguments usually have.

Some criteria were empty. A criterion is empty if it is satisfied by construction, so that it could not have failed — for instance, a condition on the Bergman exponent evaluated with a value that had been defined in terms of the domain it was meant to select. Four of the eleven were of this kind (K1889). An empty criterion contributes nothing to a probability and a great deal to its appearance.

Some criteria were the same criterion. Rank two and complex dimension five were counted separately; but at complex dimension five the classification contains two domains —  $I_{1,5}$  (the ball, rank one) and  $IV_5$  — so once the dimension is fixed the rank criterion has one candidate to choose from. (An earlier draft of this sentence said three; it counted  $I_{1,5}$  and  $I_{5,1}$  twice, which is exactly the double-counting the exceptional-isomorphism control exists to prevent. Cal caught it.) Several criteria were evaluated on a set that earlier criteria had already reduced to a singleton. Multiplying their probabilities together is the mistake the null model warns you against, made inside the null model.

And a probability was the wrong instrument. A referee does not want to hear that the alternative is unlikely. A referee wants an enumeration — every case listed, every case checked — and a theorem that says the enumeration is complete. Cartan's classification *is* such an enumeration. We should have used it as one from the start.

### What survived: one leg, one integer

Cartan's irreducible bounded symmetric domains come in four infinite families ( $I_{p,q}$ ,  $II_n$ ,  $III_n$ ,  $IV_n$ ) and two exceptional domains ( $E_6$ ,  $E_7$  types). Each carries a pair of integers, its characteristic multiplicities  $(a, b)$ ; relative to a Jordan frame  $\gamma_1, \dots, \gamma_r$  the restricted root system is  $C_r$  or  $BC_r$ , with multiplicities 1 on  $\pm\gamma_j$ ,  $a$  on  $\frac{1}{2}(\pm\gamma_i \pm \gamma_j)$ , and  $2b$  on  $\pm\frac{1}{2}\gamma_j$  — equivalently, and more simply,  $a = \dim_{\mathbb{C}} V_{ij}$  for  $i \neq j$  in the Peirce decomposition (Lyra pinned the multiplicities in Round 142; an earlier draft had  $b$  where  $2b$  belongs — vacuous here since  $b = 0$ , but wrong as a definition). Across the classification,  $a$  takes the values: 2 for type I, 4 for type II, 1 for type III,  $n - 2$  for type  $IV_n$ , 6 and 8 for the two exceptions.

*Theorem. Among irreducible bounded symmetric domains of rank at least two,  $a = 3$  has exactly one solution:  $D_{IV}^5$ .*

The proof is reading the table: only the type IV family reaches 3, and it does so at  $n = 5$  alone. We verified it three ways — a table compiled from the standard references (K1889), the same table rebuilt from root data with no reference consulted, across 66 dimensions (Elie, toy 5750), and a control that checks the low-dimensional exceptional isomorphisms where families coincide, so that no domain is counted twice or missed (K1890-PRE). The restriction to rank at least two is real: rank-one domains are the balls, and with a single  $e_1$  there are no roots  $\frac{1}{2}(e_i - e_j)$  at all, so  $a$  is not defined for them in the same sense; we state the scope rather than hide it.

Compare what the other obvious criteria leave: rank = 2 alone leaves infinitely many domains ( $I_{2,q}$  and  $IV_n$  are unbounded families; 26 under K1889's dimension cap); complex dimension 5 alone leaves two;  $a = 3$  alone leaves one. One leg, one integer.

## Where the measurement enters

Quarks come in three colours. That is a fact about the world, established at accelerators, owing nothing to any geometry. Selecting the object costs one identification and one datum, and they are different kinds of thing. The identification —

*The number of colours is the characteristic multiplicity of the domain*

— is a dictionary entry: not measured, posited, and this lecture proves the geometry cannot supply it. The datum — three colours — is measured, and it is the only number the selection consumes. Neither is a knob; together they are the price of the object. Given both, the theorem picks out  $D_{IV}^5$  and nothing else. Without the identification, the theorem picks out  $D_{IV}^5$  as the unique domain with  $a = 3$ , which is true and means nothing for physics. (Two versions of this paragraph were written in Round 142 — the shorter one said “one input”; this one counts what the selection uses and leaves the program’s downstream inputs — the sector choice of Lecture 4, the observer input of Lecture 9, the mass scale — to the lectures that name them. Casey decides what the program says about itself.)

We used to write that the geometry has “zero dimensionless free parameters.” We no longer write that. The honest count *for the choice of object* is one measured integer, named, and this is the sentence that names it. (The program as a whole has further inputs downstream — masses, a mixing corner, a CP phase — and Lecture 10 counts them; nothing here says otherwise.) (You will meet the same input in other clothes —  $n_C = \text{rank} + N_C$ , “three generations equal rank plus one” — and it is important not to count it twice. Once  $a = 3$  fixes the domain, rank 2 and dimension 5 follow; a consequence that checks out is a check, not a second leg.)

## What the uniqueness is worth — the look-elsewhere price, paid

A referee will ask the right question: how many equally natural stories — an invariant read off root data, a small target read off the world — would *also* have selected a unique domain? Elie priced it in Round 142 (toy 5753, pre-registered, labels merged so that the low-dimensional coincidences  $IV_3 = III_2$ ,  $IV_4 = I_{2,2}$ ,  $IV_6 = II_4$  count once). Six invariants, eight named targets, forty-eight stories: seven select a unique domain stably (the answer does not change when the dimension cap is raised from 30 to 400), landing on six distinct domains, and  $D_{IV}^5$  is selected by exactly one of them — this one.

And the finding that matters more, from the control: the invariant  $a$  selects a unique domain for 25 of the 29 integer targets from 2 to 30 — every target except 2, 4, 6, 8, the values the infinite families and the exceptions occupy. So the number three is not special to the invariant; the invariant is special. Had quarks come in five colours, or seven, the same sentence would have picked  $D_{IV}^7$  or  $D_{IV}^9$  with the same theorem behind it. That is the honest weight of “one leg, one integer”: the leg is always there, and the measured integer does all the selecting. The uniqueness theorem is true, and it privileges nothing about 3.

## Why we cannot yet remove it — and why that is a result

Could the identification itself be derived? Could the geometry tell us *why* colour should be its multiplicity? We asked, and the answer is a theorem of the negative kind.

The multiplicity  $a = 3$  has a concrete home inside the domain. Take a minimal tripotent — the simplest element of the Jordan triple that describes the domain — and decompose the tangent space around it. It splits as  $1 + 3 + 1$ : the middle piece, of complex dimension  $a = 3$ , is the space that used to be called the “mediator” in our earlier papers (T2543). So the multiplicity and the mediator space are one object seen twice (K1890). If colour lived anywhere in the geometry, it would live there.

Now ask what group the geometry actually supplies on that three-dimensional space. Compute it (Elie, toy 5751): it is  $U(1) \cdot SO(3)$ , of dimension four. The colour group  $SU(3)$  has dimension eight. It does not embed — the inclusion runs the other way: the geometry’s group is a proper subgroup of colour’s, and the representation it puts on the space is the complexified vector of  $SO(3)$ , which is exactly what the colour triplet becomes when restricted to that  $SO(3)$ . That is the obstruction, and it is one obstruction. (Cal’s ruling of 09-09 counted the reality type —  $3 \cong \bar{3}$  for the geometry’s triplet,  $3 \not\cong \bar{3}$  for colour’s — as a second, independent wall; Lyra’s fact read of Round 142 shows the two Frobenius–Schur indicators belong to representations of two different groups, and a complex-type representation always restricts to a real-type one on a real form, so the comparison describes the gap rather than adding to it. Cal withdrew the second obstruction the same afternoon — Section 950, calibration #40 — so the count is one, ruled.) What the geometry lacks is the complex structure that tells 3 from  $\bar{3}$ : *it can count to three but cannot orient the count* — quark from antiquark. Nor is there an escape by enlarging the group. A subgroup that acts on the colour slot must preserve it; the slot  $V_{12}$  is nondegenerate for the bilinear form, so its stabiliser in  $O(5, \mathbb{C})$  is  $O(2, \mathbb{C}) \times O(3, \mathbb{C})$ , and the image on  $V_{12}$  lies in  $\mathbb{C}^* \cdot O(3, \mathbb{C})$ , whose compact subgroups are conjugate into  $U(1) \cdot O(3)$ , dimension four. (Cal Section 946 stated this as “any compact subgroup of the full structure group is conjugate into the same dimension-4 maximal compact,” which is loose — the full structure group’s maximal compact is eleven-dimensional; the *stabiliser of the slot* is the right object, and Cal repaired his own clause in Round 142.) No group anywhere in this geometry acts on  $V_{12}$  as colour.

So the identification is not merely un-mechanised. It is *unmechanisable within the geometry*. In the falsifier register’s own vocabulary this is a floor — a theorem on the wall plus a door with a name — and it belongs in the register’s Section F as row F4 (Cal C3; the row is owed). A cleanly closed door is a result: whoever wants to derive the input must bring something the geometry does not contain, and now knows exactly what shape that something must have — a complex three-dimensional action that the tripotent decomposition does not supply.

This also touched two old rows, and we state their status exactly. T2543 said the mediator space “is colour”; that clause was *ruled* struck on 2026-09-09 (Cal Sections 944–946) — and as of this writing the registry row still carries it, because the

edit was never made; a ruling that is not swept is not a correction, and the edit is owed. T2545 argued the  $(3, 1)$  signature of spacetime from the same space: right conclusion, and the reason was re-grounded on 2026-09-11 (Grace, Round 142 G1, commit 704d402d) —  $V_{12} \cong \mathbb{C}^3$  is the complexified  $SO(3)$  vector, irreducible over  $\mathbb{C}$  and of Frobenius–Schur real type,  $3 \cong \bar{3}$ , hence not any  $SU(3)$  fundamental.

## Tier line

- Derived (classical):  $a = 3$  selects  $D_{IV}^5$  uniquely among irreducible domains of rank  $\geq 2$ . Three verifications, one instrument retained.
- Identified — the one dimensionless input to the choice of the object: colour count = characteristic multiplicity.
- Proved absent: any mechanism inside the geometry for that identification (dimension  $4 \neq 8$ ;  $3 \cong \bar{3}$ ; no larger compact subgroup).
- Not claimed: forcing; zero free parameters; “the unique geometry that can support physics”; any probability; that 3 is special to the invariant (it is not — toy 5753).
- Retired with this lecture: the eleven-criterion theorem and its null model; “the genus is 7”; “ $SU(3)$  from  $N_c = 3$ .” Owed: the registry edit striking T2543’s clause.

## What would make this lecture wrong

The theorem is a reading of a classical table; it fails only if the table is wrong, and you can rebuild the table from root data yourself (toy 5750). The identification fails if colour turns out not to be three — it is three. The absence theorem is a floor: its door is a group from *outside* the geometry acting on  $V_{12}$  by the complex fundamental, and whoever brings one replaces the identification rather than supports it. So this lecture has no live laboratory falsifier of its own. Its honest exposure is different: it *costs* the program a measured input, and every later lecture that leans on  $N_c = 3$  inherits that cost. We would rather you know the price than admire the purchase.

## Where to look

K1889 and its retained instrument for the table; Elie’s 5750 (root data) and 5751 (the group on  $V_{12}$ ); Cal’s Sections 939–946 for the ruling and the taxonomy of *selector / check / consequence* that it produced; T2543 and T2545 in the registry with their September annotations; Faraut–Korányi for the multiplicities; the legacy chapter Vol 0 Ch 9 for what we used to say, kept as the record.